

Assignment 02

For this assignment you will be writing 4 programs which should be saved independently as their own ".py" files. The filename you should use for each program is outlined in the sections below. Comment your source code and describe your code to someone who may be viewing it for the first time. Also include your name, the date, your class section and the name of your program at the top of your file.

Problem #0: Lottery Winnings Calculator (Warm-up)

You and your friends just won the lottery (congratulations!) Now you need to figure out how much money everyone is entitled to.

To do this you will be writing a short program that prompts the user for the amount they won, the number of people who are splitting the winnings, and the tax rate that will be applied to each person's share. Your program should then figure out how much each person will earn after taxes are applied to each share.

INPUT: Your program accepts the total amount won, the number of people who are splitting the winnings, and the tax rate (as an integer - i.e. 35 = 35%). Ensure that you store your data using variables of the appropriate data type.

OUTPUT: Compute the total amount won, the amount of each share, the tax due on each share and the final take-home amount for each person. Ensure that each of these values is formatted to two decimal places, and that the "comma" separator is included in your currency values, if necessary.

Sample input/output (user input is underlined):

```
How much money did you win? 1000000
How many people are splitting the winnings? 4
What is the tax rate on winnings? (i.e. 25 = 25%): 25

In total you won $1,000,000.00
Split 4 ways that amounts to $250,000.00 per person
Tax per person: $62,500.00
Take home amount per person: $187,500.00
```

This program should be named as follows: `LastNameFirstName_assign2_problem0.py`

Problem 0 Points (15)

- User input is correctly utilized to get numbers, including a whole number for the percent (5 points)

- Output lines are printed with correct formatting - \$ signs and 2 decimal places for numbers – and using input variables. Numbers are correctly calculated, either in print statements or in an intermediate step. (10 points; 2.5 points each)

Problem #1: Special Numbers

Write a program that prompts the user to enter two numbers. The numbers will always be ≥ 1 and ≤ 999 . You can assume that the user will always supply you with an integer within this range and do not need to "test" this number to ensure that it falls within the range.

Next, you will be performing a series of computations that will require you to isolate specific digits in each number based on their place value. You will also be computing a "special number" which is the result of a series of additional calculations using these values.

Specifically, your first task is to isolate the ones places from the numbers and get their product. Next, isolate the tens places of the two numbers and get their absolute difference (hint: use the `abs` function to compute the absolute value of an expression, if necessary). Then, isolate their hundreds places and get their sum. Finally, join (i.e. concatenate) the three results together to get the "special number." The sample output below shows how this program will work (user input is underlined). Make sure your program is formatted the same way as in the example:

```
Enter a number between 1 and 999: 789
Enter another number between 1 and 999: 456
Computing special number...
1. Product of ones: 9 * 6 = 54
2. Absolute difference of tens: |8 - 5| = 3
3. Sum of hundreds: 7 + 4 = 11
4. Join the numbers: 54311
Your special number is 54311!
```

Some hints to get you started: Think about how you isolate the "ones" digit - is there a math operation that you can use to extract this information? What happens when you divide the number by 10? Does that give you any usable information? How about the remainder of this division operation? Once you isolate the "ones" digit you then need to move on to the "tens" digit.

Here are some additional examples of the program:

```
Enter a number between 1 and 999: 123
Enter another number between 1 and 999: 536
Computing special number...
1. Product of ones: 3 * 8 = 18
2. Absolute difference of tens: |2 - 3| = 1
3. Sum of hundreds: 1 + 5 = 6
```

4. Join the numbers: 1816

Your special number is 1816!

Enter a number between 1 and 999: 1

Enter another number between 1 and 999: 999

Computing special number...

1. Product of ones: $1 * 9 = 9$

2. Absolute difference of tens: $|0 - 9| = 9$

3. Sum of hundreds: $0 + 9 = 9$

4. Join the numbers: 999

Your special number is 999!

This program should be named as follows: This program should be named as follows: `LastNameFirstName_assign2_problem1.py`

Problem 1 Points (27)

- User input is properly parsed to extract the ones, tens, and hundreds place from each number, using some form of the modulo operator and/or division (12 points; 4 points each)
- Output format is generally followed – i.e. for the first line, “Sum of ones,” the extracted ones from both numbers, and their sum would all be included in the print statement (3 points)
- Mathematical calculations are done correctly (9 points; 3 points each)
- Numbers are concatenated in step 4 to get the special number (3 points)

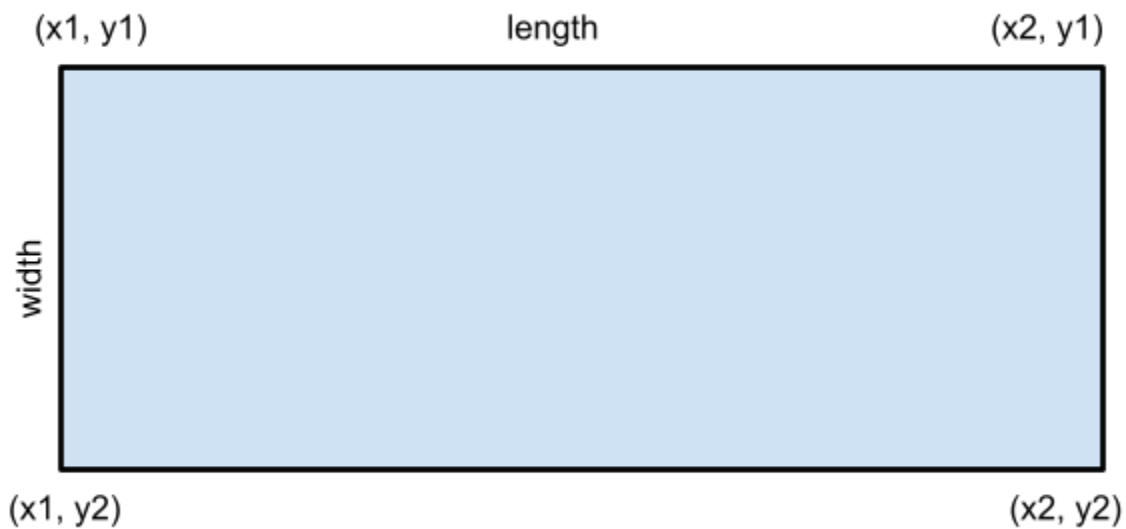
Problem #2: Your New Yard

You are redoing your yard and are buying some fence to put around it. To do so, you need to gather some information to calculate its perimeter. Your yard is a rectangle, so you will need to know the length and width to get this number.

Here's how your program should work:

- Ask the user for the first x component
- Ask the user for the second x component
- Ask the user for the first y component
- Ask the user for the second y component
- Compute the length as the absolute difference of your x components (i.e. use the `abs` function and subtract `x2` from `x1`) and print it as a float
- Compute the width as the absolute difference of your y components (i.e. use the `abs` function and subtract `y2` from `y1`) and print it as a float

- Calculate the perimeter of the rectangle using the formula $2 * \text{length} + 2 * \text{width}$. Print this number as an integer.



Below are a few sample runnings of your program (user input is underlined):

How much fencing will you need for your new yard?

First x: 5

Second x: 10

First y: -2

Second y: 2

Length: 5.0

Width: 4.0

Your yard's perimeter is 18

How much fencing will you need for your new yard?

First x: 3.5

Second x: 7

First y: 3

Second y: 9

Length: 3.5

Width: 6.0

Your yard's perimeter is 19

This program should be named as follows: `LastNameFirstName_assign2_problem2.py`

Problem 2 Points (25)

- Program correctly utilizes user input to get 2 x points and 2 y points and converts to numbers (8 points; 2 points each)
- Length is correctly calculated regardless of input and is a float (5 points)
- Width is correctly calculated regardless of input and is a float (5 points)
- Perimeter is correctly calculated and is an integer (7 points)

Problem #3: Variable Rate Loan

You are taking out a loan and are interested in looking at how this amount will grow with a variable interest rate. In this case, the interest rate will change every month. Usually, this change occurs due to market conditions, but in this case, the new rate will be randomly selected by Python.

For this problem, you will be writing a program that calculates how much a person will owe each month after accounting for the monthly increase on the loan due to the variable interest rate. This program should ask the user for an initial loan amount, a lower bound for the interest rate, and an upper bound for the interest rate.

The program will randomly choose a different integer in the bounds given as a variable interest rate for each month. The program will then generate a 3-month projection that illustrates how much money the user can expect to owe.

Here's a sample running of the program:

This program will project how much a variable interest-rate loan will grow over a 3-month period.

To begin, enter how much money you would like to initially borrow (i.e. 20000): 20000

Next, enter a lower bound for the monthly interest rate. For example, enter 5 for 5%: 3

Finally, enter a lower bound for the monthly interest rate. For example, enter 10 for 10%: 12

----- Calculating -----

Month	Starting Balance	Interest Rate	Ending Balance
1	\$20,000.00	11%	\$22,200.00
2	\$22,200.00	5%	\$23,310.00

3	\$23,310.00	8%	\$25,174.80
---	-------------	----	-------------

You can assume that the user will enter reasonable input values (i.e. all inputted values will be valid positive floating point numbers.) Ensure that your program is formatted correctly so that your values line up as in the sample output above (hint: use the `format()` function.) Make sure output is formatted with proper commas, dollar signs, and percent signs, and that decimal points match those above (2 for money, none for percentages).

This program should be named as follows: `LastNameFirstName_assign2_problem3.py`

Problem 3 Points (33)

- Input is correctly utilized, including a whole number being used for the percent (6 points)
- The random function is used to get 3 random interest rates (6 points)
- Output is correctly lined up into columns (6 points)
- Numbers are correctly formatted – integers for interest rates with % signs, commas and \$ with 2 decimal points for balances (6 points)